

An Improved Two-Scale Model for Sea Surface Scattering in the Transition Range of Incidence Angles

Jing Ye^{ID}, Yong Wan^{ID}, *Member, IEEE*, Chenqing Fan, and Yongshou Dai

Abstract—An improved two-scale model (I-TSM) for the incidence angle transition range (5° – 11°) is proposed by analyzing the contributions of both quasi-specular and Bragg scatterings to the total sea surface scattering. With normalized radar cross section (NRCS) products of surface wave investigation and monitoring (SWIM) instrument, the mean square slope (MSS) and the reflection coefficient of the geometrical optics (GO) model are modified to simulate the quasi-specular scattering in I-TSM, and a Bragg scattering contribution factor adaptive to the incidence angle and wind speed is fit to provide the Bragg scattering component for I-TSM. At different incidence angles and wind speeds, I-TSM combines the quasi-specular and Bragg scatterings in variable proportions by the fitting factor. I-TSM avoids the critical angle selection and separation problems of the traditional TSM in the incidence angle transition range. Compared with the GO model and traditional TSM, I-TSM has significantly higher accuracies with a root-mean-square error (RMSE) of 0.309 dB, a relative error (RE) of 2.019%, and a scattering index (SI) of 0.03 when SWIM NRCS products are the standard data in the validation. I-TSM will provide high-precision NRCS simulation data in the incidence angle transition range for scattering characteristics analysis and echo simulation of SWIM.

Index Terms—Incidence angle transition range, scattering contribution analysis, surface wave investigation and monitoring (SWIM), two-scale model (TSM).

I. INTRODUCTION

THE measurement and characteristic analysis of sea surface scattering are essential for radar signal simulation and ocean parameter inversion [1]. According to the electromagnetic scattering theories, the sea surface scattering mechanisms vary with radar incidence angles or wavelengths. Solved with the Kirchhoff approximation (KA) method,

the quasi-specular scattering from large-scale waves is dominated at low incidence angles [2]. Bragg scattering from small-scale waves dominates the sea surface scattering at medium incidence angles, commonly calculated by the small perturbation method (SPM) [3]. Although the small slope approximation (SSA) [1] and the integral equation method (IEM) [4] are of wide application ranges and high calculation accuracies, they suffer from complex algorithms and heavy consumption of time and computer memory [5]. As the sea surface is the superposition of waves on large and small scales, considering both quasi-specular and Bragg scattering mechanisms, the two-scale model (TSM) [6] is applied to a broader range of incidence angles. However, by separating the two scatterings action ranges with a critical incidence angle or cutoff wavenumber, the traditional TSM usually calculates the normalized radar scattering cross section (NRCS) of the two ranges with corresponding models, which is essentially a separation model [7], [8]. Hence, in the transition range of small and medium incidence angles where two scatterings coexist, the simulation accuracy of NRCS by the traditional TSM is generally low. Due to the sensitivity of traditional TSM to the separation point selection, the critical values set by experience are not uniform in studies, which leads to differences in model accuracy [9]–[12]. In the early satellite scatterometry, it was unsuccessful to use purely analytical models of radar scatter on the sea surface because the near-surface wind is not the only determining parameter. Different from the analytical models, the geophysical model function (GMF) is the empirical model fit by radar observation data [10]. GMF represents the quantitative relationship between NRCS, sea surface wind (speed and direction), and incidence angle, of which the fitting form lacks a theoretical basis. To solve the problems of the low accuracy of the analytical scattering model and the lack of theoretical basis of the empirical GMF, a physically based GMF needs to be developed.

An accurate GMF contributes to NRCS simulation and wind speed retrieval of the radar. Launched in October 2018, Surface Wave Investigation and Monitoring (SWIM) is a real aperture radar operating at the Ku-band (13.575 GHz). It is the first spaceborne sensor to obtain wave direction spectrum with multiaimuth and multibeam technology [13]. SWIM observes at 0° – 11° incidence angles, the NRCS of which is usually simulated by the quasi-specular model [14]–[16].

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Jing Ye is with the College of Control Science and Engineering, China University of Petroleum, Qingdao 266580, China (e-mail: yejing1115@163.com).

Yong Wan and Yongshou Dai are with the College of Oceanography and Space Informatics, China University of Petroleum, Qingdao 266580, China (e-mail: wanyong@upc.edu.cn; daiys@upc.edu.cn).

Chenqing Fan is with the First Institute of Oceanography, Ministry of Natural Resources, Qingdao 266061, China (e-mail: fanchenqing@fio.org.cn).

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However, at about 10° or even smaller incidence angle, the Bragg scattering affects the NRCS to a degree that is not negligible [17]. Therefore, the transition range of small and medium incidence angles [18] in which both the quasi-specular and Bragg scattering contribute to the total NRCS is a part of the incidence angles of SWIM. In the transition range, it is necessary to take the contributions of two scatterings into account to simulate the SWIM NRCS accurately.

This letter aims to propose an improved TSM (I-TSM) for the transition range of incidence angles, which is actually a physically based GMF. SWIM NRCS products are the data source for constructing and validating the model. Section II modifies the quasi-specular model, and Section III determines the incidence angle transition ranges at various wind speeds. By analyzing quasi-specular and Bragg scattering contributions to the total sea surface scattering at different incidence angles and wind speeds, I-TSM is proposed and verified in Section IV. The conclusion is in Section V.

II. MODIFIED QUASI-SPECULAR MODEL

An accurate quasi-specular model provides high-precision NRCS simulations at low incidence angles and plays an essential role in constructing I-TSM.

The GO is a common model for quasi-specular scattering [2]

$$\sigma_{GO}^0 = \frac{|R(0)|^2}{s^2} \cdot \sec^4 \theta \cdot \exp\left(-\frac{\tan^2 \theta}{s^2}\right) \quad (1)$$

where $R(0)$ is the Fresnel reflection coefficient, s^2 is the mean square slope (MSS) of the sea surface, and θ is the radar incidence angle. $R(0)$ and s^2 are often modified in GO [2]. First, R applies only to the surface on which there are no waves whose wavelength is equal to or smaller than the radar wavelength. For the real sea surface, small-scale waves change the reflection characteristics of large-scale waves to electromagnetic waves, only the MSS generated by the wave whose wavelength is larger than that of radar can affect the radar observed scattering, and the effective reflection coefficient R_{eff} represents the reflection characteristics of these large-scale waves. Second, s^2 represents the overall MSS of the sea surface, which is generated by fluctuations in all wavelengths. While the radar filter MSS s_{filter}^2 is only contributed by the fluctuation whose wavelength is larger than that of radar, it represents the fluctuations of sea surface waves that radar observes. In the practical application of the GO model, R_{eff} and s_{filter}^2 better reflect the sea surface observation of radar than R and s^2 , respectively.

ERA5 provides global wind field data [19]. Based on a quality assessment with data from 16 NDBC buoys, the root-mean-square errors (RMSEs) of ERA5 wind speed data are about 1.2 m/s, which refers to high quality. Thus, ERA5 wind data matched with SWIM products are used for model construction and verification in this letter. Since the scattering mechanism at the near-nadir incidence angle is considered the quasi-specular [2], SWIM NRCS products from July to September 2020 at 2° – 5° are data sources for the modification. Related to wind speed at 10 m height above the surface, U_{10} ,

TABLE I
RMSEs OF GO MODELS WITH DIFFERENT PARAMETERS

Scheme number	Reflection coefficient	MSS	RMSE of GO(dB)
1	Fresnel reflection coefficient	MSS of all the sea surface waves	2.544
2	Proposed by Li	Proposed by Jackson	1.737
3	Proposed in (2)	Proposed in (3)	0.218

R_{eff} , and s_{filter}^2 are fit by the least-square method as

$$|R_{\text{eff}}(0)|^2 = -0.1832 \exp(-1.1790 U_{10}) - 0.0100 U_{10} + 0.5130 \quad (2)$$

$$s_{\text{filter}}^2 = 0.0103 + 0.0007 U_{10}. \quad (3)$$

With SWIM NRCS products from July to September 2021, the RMSEs of the three GO models based on different reflection coefficients and MSSs are calculated at 2° – 5° . $\text{RMSE} = ((1/n) \sum_{i=1}^n (A_i - B_i)^2)^{1/2}$, where n is the data amount and A_i and B_i are NRCSs of the model and SWIM. Table I shows the accuracies of GO models based on different parameters. The model of scheme 1 has low accuracy because R and s^2 fail to reflect the actual radar observation situations. $R_{\text{eff}}(0)$ proposed by Li *et al.* [20] and s_{filter}^2 proposed by Jackson *et al.* [21] are adopted in scheme 2, and the model accuracy is still not high. Fitting data sources of the two parameters are from TOPEX/POSEIDON (T/P) altimeter and airborne radar. Due to the different operating parameters and calibration accuracy settings, there is an inevitable “system offset” between the instrument observations [22]. It leads to different NRCS measurements between instruments. Thus, the model of scheme 2 does not agree well with SWIM NRCS. In comparison, with $R_{\text{eff}}(0)$ and s_{filter}^2 fit by SWIM products, the modified GO model simulates NRCS with higher precision.

III. TRANSITION RANGE OF INCIDENCE ANGLES

In the transition range of incidence angles, the mechanism of sea surface scattering is complex. However, the transition range under the SWIM observation condition has not been studied adequately. Since the letter aims to propose a high-precision physically based GMF for simulating SWIM NRCS, it is assumed that the scattering in the transition range is the combination of the quasi-specular and Bragg mechanisms. Then, considering that the modified GO model accurately simulates the quasi-specular scattering, the specific transition ranges where Bragg scattering influences are obtained by comparing NRCSs of the modified GO model and SWIM. The comparison results are shown in Fig. 1.

Based on the above assumption, the total scattering consists of two components. With increasing incidence angle in the transition range, the contribution ratio of quasi-specular scattering decreases continuously. Accordingly, the Bragg contribution ratio increases monotonically. In this letter, the lower limit of the transition range is determined by the minimum angle satisfying the condition of a positive and monotone increasing ΔNRCS . Moreover, SWIM provides NRCS data every 0.5° in the 0° – 11° range. Due to the limitation of the SWIM incidence angles, the upper limit of

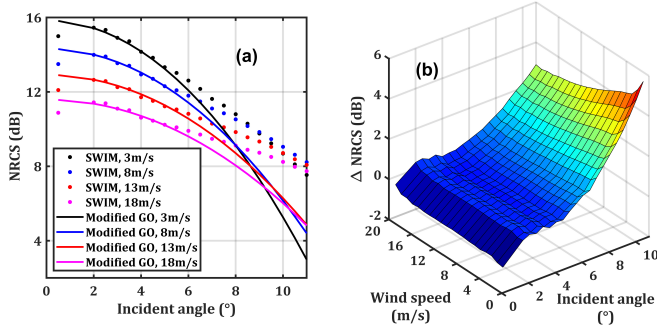


Fig. 1. Comparisons of NRCSs from SWIM and the modified GO model. (a) NRCSs. (b) Differences of NRCSs ($\Delta\text{NRCS} = \text{NRCS}_{\text{SWIM}} - \text{NRCS}_{\text{GO}}$).

TABLE II
RESULTS OF TRANSITION RANGE OF INCIDENCE ANGLES

Wind speed (m/s)	Incidence angle transition range (°)
[1, 5]	[5, 11]
[5, 20]	[5.5, 11]

the transition range is fixed at 11° here. Based on the results in Fig. 1(b) and the above determining criterion of the transition range, the specific ranges are sorted out in Table II. The minimum angle of transition range varies with wind speeds, at which Bragg scattering begins to have a nonnegligible effect on the total sea surface scattering. Table II provides specific transition ranges of incidence angles under different wind speeds for constructing the improved model in the next step.

IV. IMPROVED TSM

A. Establishment of the I-TSM

In the transition range of incidence angles, we assume that formally, the total scattering can be represented as a combination of the quasi-specular and Bragg mechanisms. However, the direct addition of the two components can not accurately describe the total. Hence, I-TSM is proposed as

$$\sigma_0^p = a \cdot \sigma_{sp} + b(U_{10}, \theta) \cdot \sigma_{br}^p \quad (4)$$

where p is the polarization state (due to the operating parameters of SWIM, only HH polarization is considered here), σ_0^p is the NRCS of the total sea surface scattering, σ_{sp} is the NRCS of quasi-specular scattering calculated by the modified GO model (1), σ_{br}^p is the NRCS of Bragg scattering calculated by the corresponding model, and a and $b(U_{10}, \theta)$ are the contributing factors of quasi-specular and Bragg components to the total scattering, respectively. In this letter, a is fixed at 1 as the quasi-specular scattering is dominant within the incidence angle range of SWIM (0° – 11°). Bragg scattering is a supplementary contributor to the total scattering in the transition range, and $b(U_{10}, \theta)$ is jointly determined by wind speeds and incidence angles.

The composite Bragg model [23] describes σ_{br}^p from small-scale waves influenced by large-scale waves on the sea surface

$$\sigma_{br}^p(\theta, \varphi) = \sigma_{0br}^p(\theta, \varphi) \left(1 + g^p \overline{s_i^2} - \frac{M_{i0}^p}{B_{r0}} \overline{s_i \tilde{B}_r} \right) \quad (5)$$

where $\sigma_{0br}^p(\theta, \varphi) = 16\pi k_r^4 |G_p(\theta)|^2 F_r(\varphi, k_{br})$ is the NRCS for a pure Bragg process, φ is the azimuth of the radar antenna, $k_{br} = 2k_r \sin \theta$ is the Bragg wavenumber,

k_r is the radar wavenumber, and the folded spectrum $F_r(\varphi, k)$ is related to the directional wavenumber spectrum $F(\varphi, k)$ (Elfouhaily spectrum [24] is used) by $F_r(\varphi, k) = 0.5(F(\varphi, k) + F(\varphi + \pi, k))$. In addition, G_p and g^p are polarization scattering coefficients in the pure and composite Bragg models: $|G_h(\theta)|^2 = \frac{\cos^4 \theta}{(0.111 \cos \theta + 1)^4}$, $|G_v(\theta)|^2 = \frac{\cos^4 \theta (1 + \sin^2 \theta)^2}{(\cos \theta + 0.111)^4}$, $g_h = \frac{1}{2\sigma_{0br}^h} \frac{\partial^2 \sigma_{0br}^h}{\partial \theta^2} + \frac{2}{\sin^2 \theta} \frac{G_v}{G_h} \frac{\overline{s_i^2}}{s_i^2}$, and s_i^2 and $\overline{s_i^2}$ are MSSs in the directions of the incidence plane and its vertical, related to the upwind MSS $\overline{s_u^2}$ and crosswind MSS $\overline{s_c^2}$, respectively. The ratio of $\overline{s_c^2}$ to $\overline{s_u^2}$ is about 0.8 [25], and the sum is the total sea surface MSS. $((M_{i0}^p)/B_{r0})\overline{s_i \tilde{B}_r}$ is the cross correlation of the tilt and hydrodynamic modulations [23], the value of which is minimal compared with other items in (5), so it is not considered here.

B. Fitting Factor of Bragg Scattering

For the consistency between model values and SWIM measurements, it is necessary to fit the Bragg scattering factor $b(U_{10}, \theta)$ of I-TSM under various incidence angles and wind speeds. SWIM NRCS products from July to September 2020 are the data source for total scattering σ_0^p in (4), and NRCSs from the modified GO and the composite Bragg models are calculated by (1)–(3) and (5). Then, the raw Bragg scattering contribution factors b_{raw} are calculated by (4) under various combinations of wind speeds (a range of 1–20 m/s and an interval of 1 m/s) and incidence angles (a range of 5° – 11° and an interval of 0.5°). Finally, $b(U_{10}, \theta)$ is fit based on b_{raw} by the BP neural network method [26]. In detail, the node number settings for the input, hidden, and output layers are 2, 10, and 1, respectively. The transfer functions are “tansig” for the hidden layer and “purelin” for the output layer, and the training method is “trainlm”.

From the fit $b(U_{10}, \theta)$ in Fig. 2, there are apparent distribution characteristics that the Bragg contribution to the total enhances with increasing incidence angles and decreasing wind speeds. Fitting accuracies are calculated by the RMSE and the correlation coefficient $r = \frac{\text{Cov}(b_{\text{raw}}, b(U_{10}, \theta))}{\sqrt{\text{Var}[b_{\text{raw}}] \text{Var}[b(U_{10}, \theta)]}}$, where $\text{Cov}(b_{\text{raw}}, b(U_{10}, \theta))$ is the covariance of data before and after fitting and $\text{Var}[b_{\text{raw}}]$ and $\text{Var}[b(U_{10}, \theta)]$ are variances of data before and after fitting, respectively. In addition to r of the overall data, the correlation coefficient r^* of data under low winds (1–4 m/s) and high incidence angles (8° – 11°) is calculated to verify the fit $b(U_{10}, \theta)$ with large values. In Fig. 2, the fitting accuracy of $b(U_{10}, \theta)$ is high so that it can be adopted in I-TSM.

C. Comparisons Between Models and SWIM Measurements

In traditional TSMs, the critical incidence angles for separating quasi-specular and Bragg scatterings are usually greater than 10° by empirical selection. Therefore, the traditional TSM is considered equivalent to the quasi-specular model in the incidence angle range of SWIM (0° – 11°). In view of this, the modified GO model in Section II is selected for I-TSM comparison in validation.

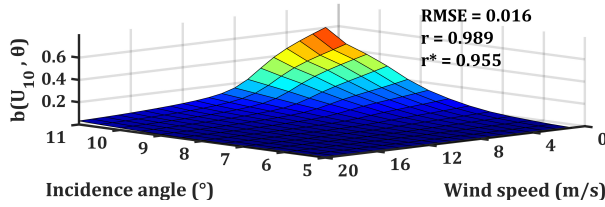
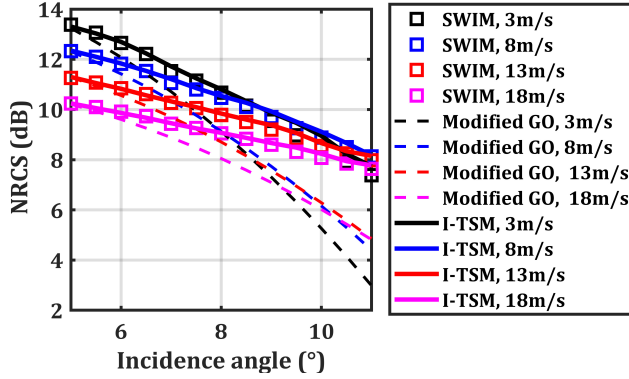
Fig. 2. Fit $b(U_{10}, \theta)$.

Fig. 3. NRCSs of the models and SWIM in the incidence angle transition range.

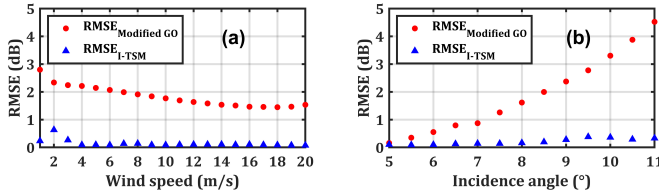


Fig. 4. RMSEs of the models in different conditions. RMSEs at different (a) wind speeds and (b) incidence angles.

The average values of SWIM NRCS products in various combinations of wind speeds (a range of 1–20 m/s and an interval of 1 m/s) and incidence angles (a range of 5°–11° and an interval of 0.5°) from July to September 2021 are the standard data in the validation. Fig. 3 shows the NRCSs of models and SWIM in the incidence angle transition range under four typical wind speeds. NRCS differences between the modified GO model and SWIM increase significantly with increasing angles. The consistency between I-TSM and SWIM is high, benefitting from the complement of the Bragg scattering contribution in the transition range. RMSEs of models at various wind speeds and incidence angles are analyzed as well. As shown in Fig. 4, when the wind speed or incidence angle increases, RMSEs of the modified GO model have an entirely opposite numerical variation trend. It illustrates that the Bragg scattering has a greater influence on the total surface scattering under lower wind speeds and higher incidence angles. This result is identical to the distribution characteristics of $b(U_{10}, \theta)$ in Fig. 2, and the principle can be explained by the scattering observation process of SWIM. The fluctuation of the large-scale wave is slight at low wind speed, and the quasi-specular scattering of large-scale waves decreases rapidly with the gradually increasing near-nadir incidence angle. In this process, the contribution of Bragg scattering to the total increases markedly, which is caused by the small-scale waves superimposed on the large-scale ones.

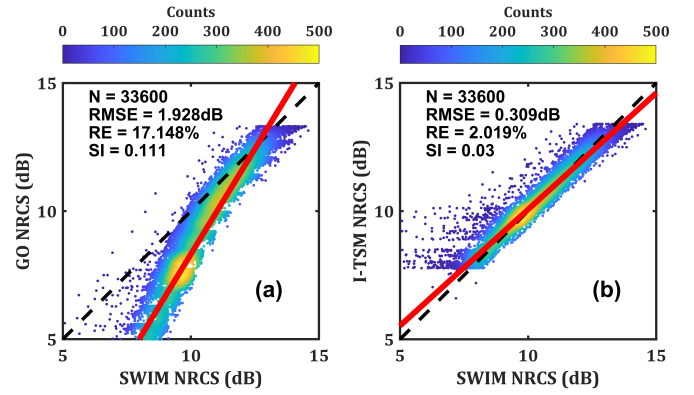


Fig. 5. Scatterplots for NRCS comparisons between the model and SWIM measurements. (a) Modified GO model and SWIM measurements. (b) I-TSM and SWIM measurements.

Thus, the modified GO model is of low accuracy under low wind speed and high incidence angle in the transition range. Fortunately, the advantage of I-TSM over the GO model is that the contribution of composite Bragg scattering can be increased by a fitting factor $b(U_{10}, \theta)$ to compensate for the lack of quasi-specular scattering. As a result, accuracies of I-TSM under various conditions in Fig. 4(a) and (b) are all significantly higher than those of the modified GO model.

The above validation is based on the average data of SWIM NRCS products at various combinations of wind speed and incidence angle. In fact, when observing the sea surface with SWIM, the angle of incidence is not constant and the state of the sea varies across the surface. Therefore, to further test I-TSM in real sea conditions, SWIM NRCS products as of July 1, 2021 are selected in the range of incidence angles of 5°–11° as standard data, and the corresponding ERA5 wind field data are used to make NRCS comparisons between models and SWIM measurements. The accuracy indexes are RMSE, relative error (RE) ($RE = \frac{1}{n} \sum_{i=1}^n \left(\frac{|A_i - B_i|}{B_i} \right) \times 100$), and scattering index (SI) ($SI = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n [(A_i - \bar{A}) - (B_i - \bar{B})]^2}}{\bar{B}}$), where $\bar{A}$ and $\bar{B}$ represent the average NRCSs of the model and SWIM, respectively. From Fig. 5(a), the modified GO model underestimates most of the observed data, and it can be inferred that the main reason is the neglect of Bragg scattering in the transition range. RMSE, RE, and SI of I-TSM in Fig. 5(b) are significantly lower than those of the modified GO model within 33 600 points. These accuracy results further prove that NRCS calculated by I-TSM is highly consistent with SWIM measurements. I-TSM has an obvious advantage in simulating NRCS in the transition range of incidence angles over the GO model and the traditional TSM.

V. CONCLUSION

First, in the GO model, the reflection coefficient and MSS are modified based on the NRCS obtained from SWIM to improve the accuracy of quasi-specular scattering at near-nadir incidence angles. Then, by analyzing the NRCS differences between the modified GO model and the SWIM products at 0°–11°, the transition ranges of incidence angles at different

wind speeds are determined. Finally, an improved I-TSM model for the transition range is proposed with a fitting Bragg scattering parameter. This parameter depends on the wind speed near the sea surface and the incidence angle of the radar. Thus, I-TSM has a significant advantage over traditional TSM, which implements a single division parameter in the spectrum of surface heights. In a sense, I-TSM is a physically based GMF rather than a rigorous scattering model. Model validation shows that I-TSM reproduces NRCS SWIM measurements with greater accuracy than the GO model or traditional TSM. The proposed approach can help develop geophysical modeling functions based on physical scattering models applicable to other radar measurements. Future studies will consider other important wave factors, such as non-Gaussian effects and wave breaking, to further improve the accuracy of the model and expand its range of applicability.

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